Inorganic CUME

March-8-2013

The newly established "Energy Storage and Battery Hub" led by Argonne National Lab under a $200 million grant, the highest ever from the US-DOE, implies that "energy storage and battery" will be a hot topic in research, economy, and job market over decades to come.

(1). Please pick a primary (not rechargeable) battery of your choice, and briefly discuss the inorganic reactions on the anode and cathode during the battery's charging and discharging processes. (**20'**)

(2). Please pick a secondary (rechargeable) battery of your choice to discuss the inorganic reactions in the charging and discharging processes at the anode and cathode of the battery. (**20'**)

(3). Manganese oxides are commonly used as battery materials. Briefly define the EMF. Indicate the sign (positive or negative) of each EMF in the following diagram and briefly explain who so, based on the fact that the oxidation states (IV) and (II) being more stable than the others. (**20'**)

$$Mn^{4+} \xrightarrow{\text{EMF:}} Mn^{3+} \xrightarrow{\text{EMF:}} Mn^{2+} \xrightarrow{\text{EMF:}} Mn^{0}$$

(4). A classic type of battery in most cars uses too much of an element that is heavy and toxic. Propose a new electrode material for largely replacing the toxic element in this type of battery, and briefly discuss your inorganic chemistry for supporting your proposal. (**20'**)

(5). Please propose a novel inorganic reaction (or a material) for safely and economically storing energy other than the electrical energy, and briefly discuss the working principle. (**20'**)